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/*  
 * C Program to Perform Merge Sort using Recursion  
 */
```

```
#include <stdio.h>
```

```
void mergeSort(int [], int, int, int);
```

```
void partition (int [], int, int);
```

```
main()  
{  
    int a[50];  
    int i, n;  
  
    printf("Enter total number of elements:");  
    scanf("%d", &n);  
    printf("Enter the elements:\n");  
    for(i = 0; i < n; i++)  
    {  
        scanf("%d", &a[i]);  
    }  
  
    partition(a, 0, n - 1);  
    printf("After merge sort:\n");  
    for(i = 0; i < n; i++)  
    {  
        printf("%d  ", a[i]);  
    }  
}
```

```
void partition(int a[],int low,int high)  
{  
    int mid;  
    if(low < high)  
    {  
        mid = (low + high) / 2;  
        partition(a, low, mid);  
        partition(a, mid + 1, high);  
        mergeSort(a, low, mid, high);  
    }  
}
```

```

void mergeSort(int a[],int low,int mid,int high)
{
    int i, m, k, l, temp[50];
    l = low;
    i = low;
    m = mid + 1;
    while ((l <= mid) && (m <= high))
    {
        if (a[l] <= a[m])
        {
            temp[i] = a[l];
            l++;
        }
        else
        {
            temp[i] = a[m];
            m++;
        }
        i++;
    }
    if (l > mid)
    {
        for (k = m; k <= high; k++)
        {
            temp[i] = a[k];
            i++;
        }
    }
    else
    {
        for (k = l; k <= mid; k++)
        {
            temp[i] = a[k];
            i++;
        }
    }

    for (k = low; k <= high; k++)
    {
        a[k] = temp[k];
    }
}

```